

Jorge Cebola Borbinha

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Engineer and quantitative researcher with 7+ years of experience in predictive simulations, data analysis, and pipeline development. Proficient in engineering production-grade Python models and statistical modeling. Dedicated to applying advanced technical capabilities to solve complex commercial challenges and driving data-informed decision-making.

Experience

Physics and Chemistry Teacher | Ministério da Educação da República Portuguesa | 11/2025 - 01/2026

- Translated complex Physics and Chemistry concepts for diverse 7th-12th grade classes and developed individual improvement plans, increasing student engagement and comprehension across native and non-native speakers.
- Directed administrative and grading oversight for a class of students while successfully mediating interpersonal conflicts through structured negotiation and inclusive classroom management techniques.

Data Analyst | Vaidade das Folhas Lda. | 03/2025 - 11/2025

- Optimized vineyard yield and resource utilization by synthesizing multi-year agricultural datasets (e.g. weather, soil, geography) to uncover key production trends using Python and Excel.
- Isolated underperforming production cycle variables through statistical analysis and the construction of data-driven dashboards, enabling leadership to execute strategic decisions that improved agricultural outcomes.
- Coordinated with multi-institutional stakeholders to ensure full compliance with regulatory standards.

Research Fellow | Instituto Superior Técnico, Universidade de Lisboa | 03/2018 - 01/2025

- Defined technical protocols and ensured legislative compliance, alongside health and radiation protection regulatory bodies and industry stakeholders, for advanced image data acquisition and radiological field exercises.
- Accelerated the generation of complex human-like 3D voxel models and minimized data wrangling time by designing and implementing Python pipelines and scripts to process and visualize medical imaging datasets exceeding 20M datapoints.
- Eliminated manual processing workloads and exposed critical radiation dose underestimations of 20-40% by developing a FORTRAN ETL pipeline to centralize human voxel model generation and manipulate organ volumes.
- Developed Python scripts using Monte Carlo simulation data and regression analysis, and MATLAB scripts for performance analysis of radiation detectors, to locate and identify radiation sources in northern Portugal from drone scouting data.

Skills

Programming: Python (NumPy, pandas, plotly, matplotlib, seaborn, SciPy), Fortran, C++, MATLAB, R

Data Analysis: Quantitative Research Methodologies, Data Wrangling, Statistical Data Modeling and Regression, Data Visualization

Technical Tools: Monte Carlo Simulation, Excel (Advanced), Bash, Prompt Engineering (Google Gemini, Perplexity)

Communication and Adaptability: Scientific and Technical Writing, Scientific Presentation, Regulatory and Legislative Compliance, International Conference Organization

Education

Advanced Studies Diploma in Physics Engineering | Instituto Superior Técnico, Universidade de Lisboa | 2023

Master of Science in Biomedical Engineering | Faculdade de Ciências e Tecnologia, Universidade NOVA de Lisboa | 2017

Certification: Microsoft Professional Certificate DEV210x: Introduction to C++ | 2018

Honors and Awards

European Radiation Dosimetry Group Young Scientist Award | 2023

Languages

Portuguese: C2 | English: C2 | Spanish: B2